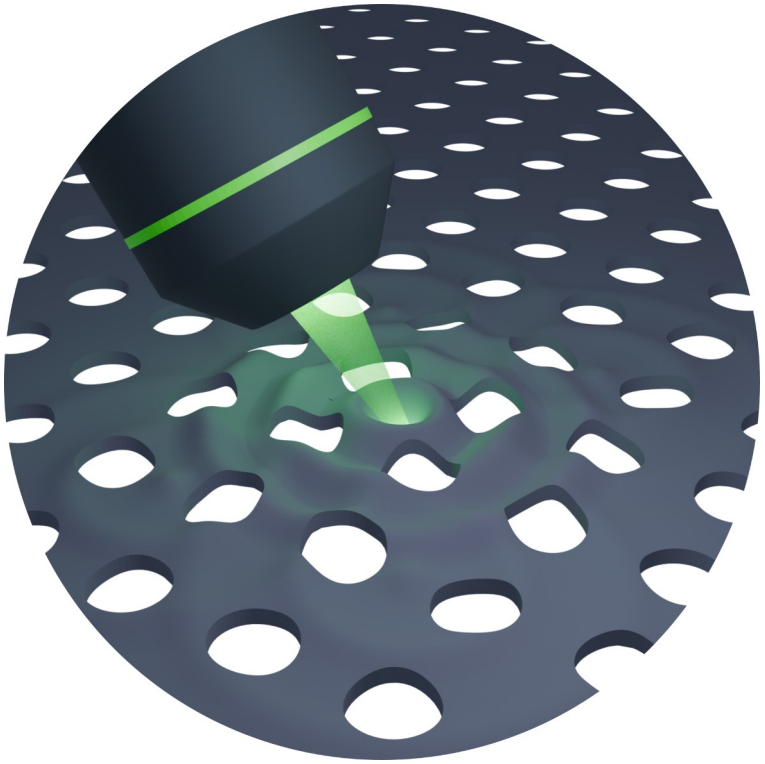


Time reversal of BLS signal?

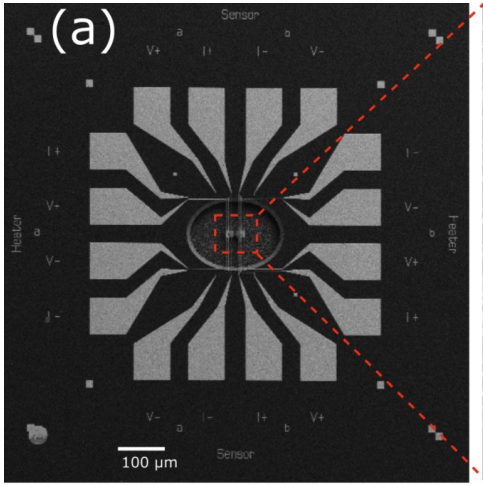
Roman Anufriev

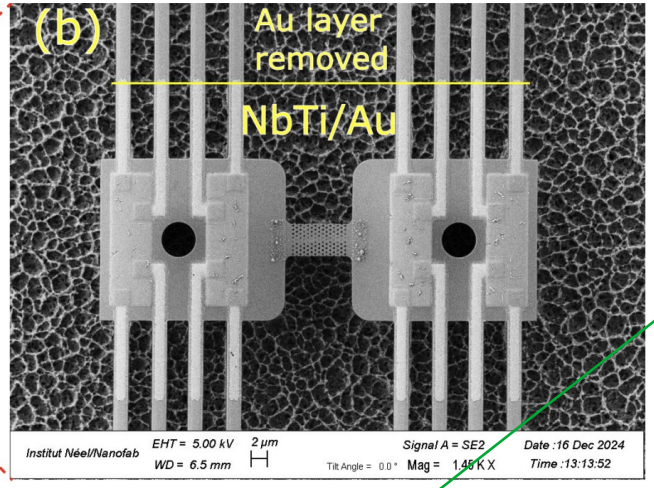
The French National Centre for Scientific Research

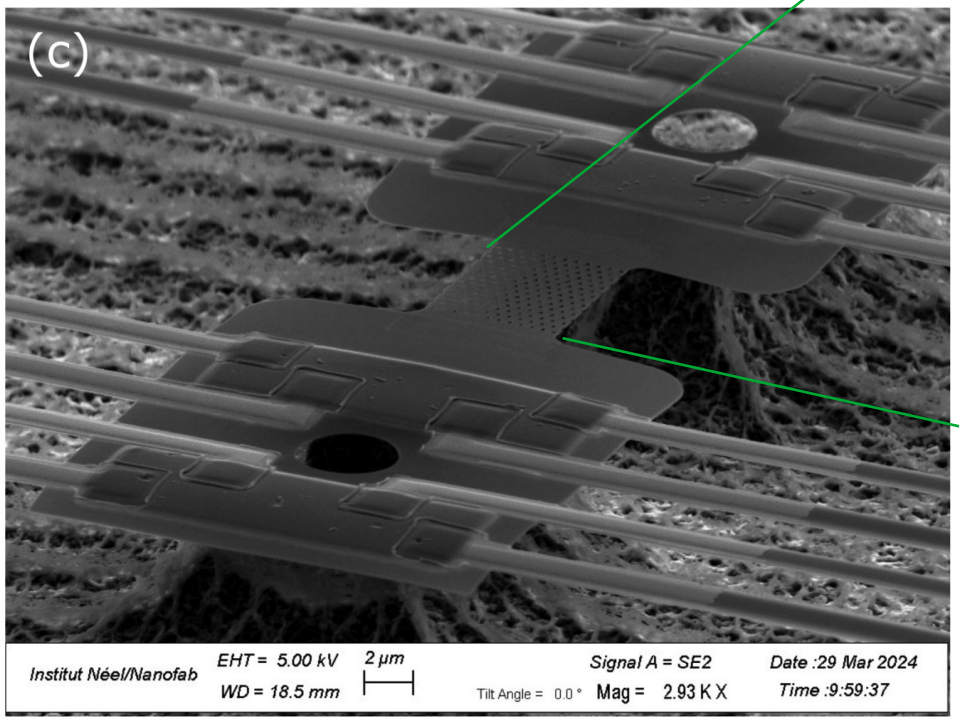
The University of Tokyo

Experiments in Grenoble

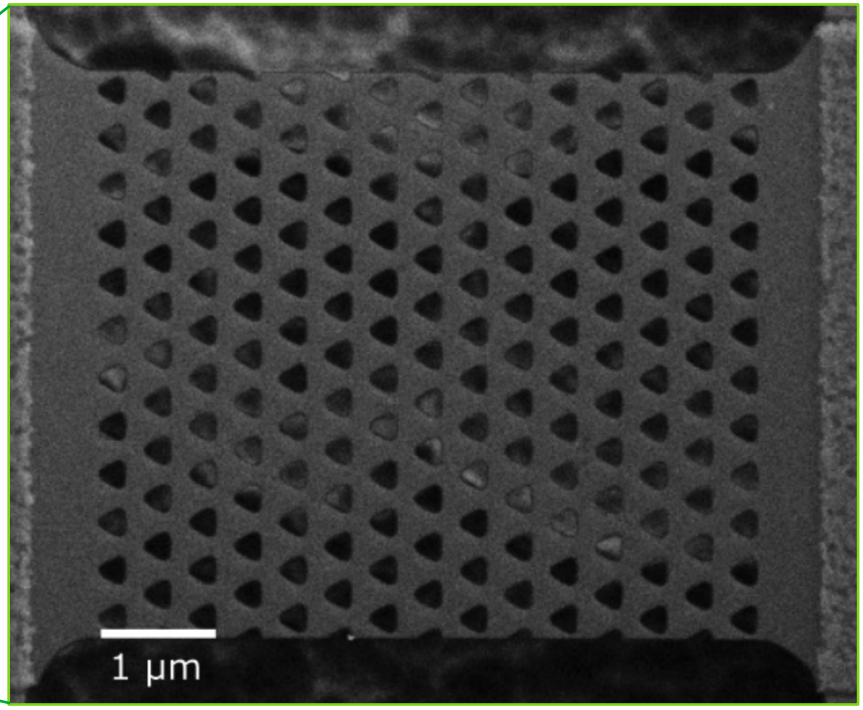
The group in Institute NEEL measures phononic structures at ultra-low temperatures:







Forward

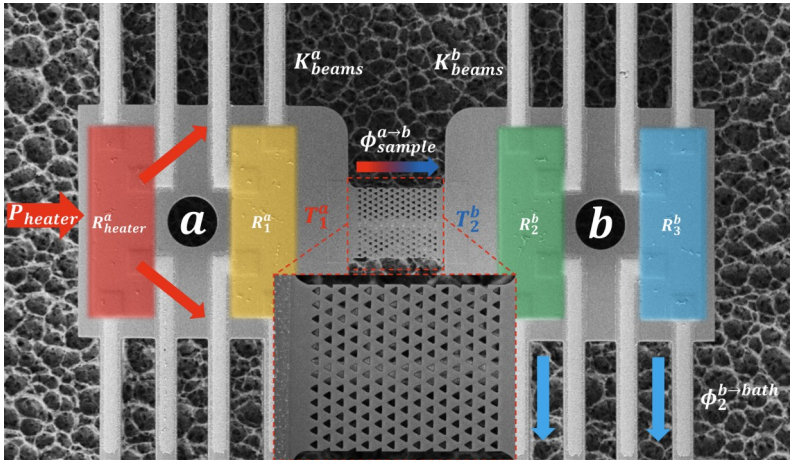


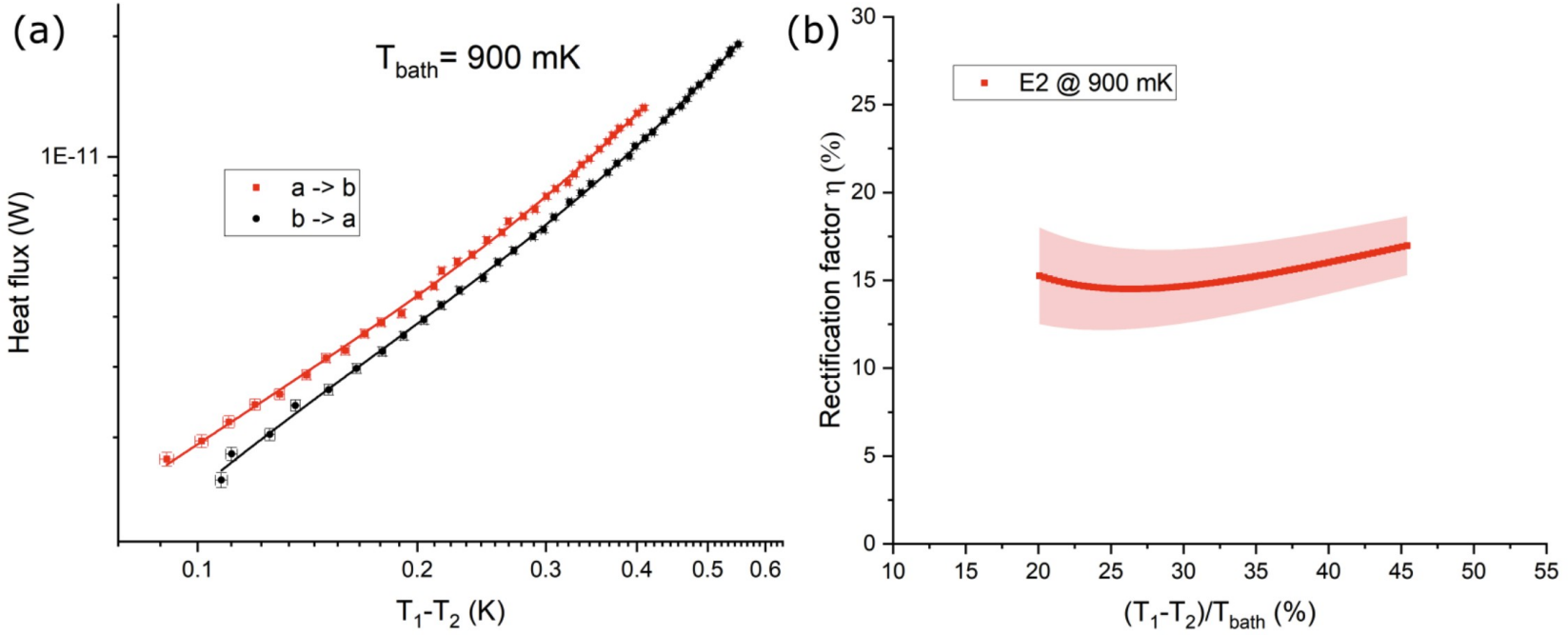
Backward

Boris Brisuda, PhD Thesis (2025)

Experiments in Grenoble

They measured thermal rectification caused by triangular holes:





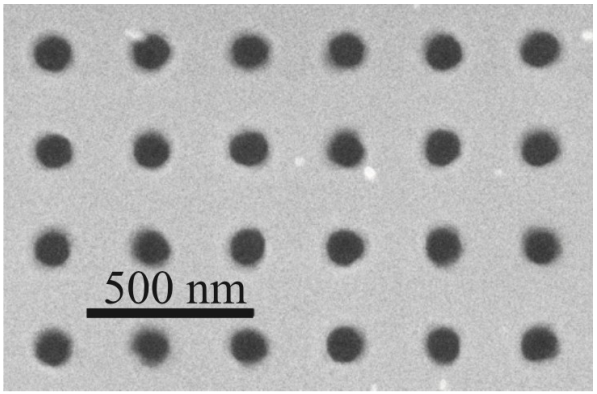
We tried many things to explain it, but so far nothing worked...

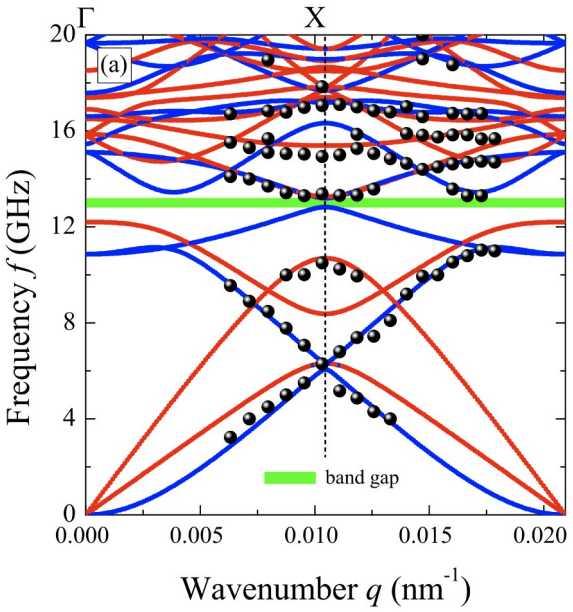
Boris Brisuda, PhD Thesis (2025)

Periodic structures cause phonon interference

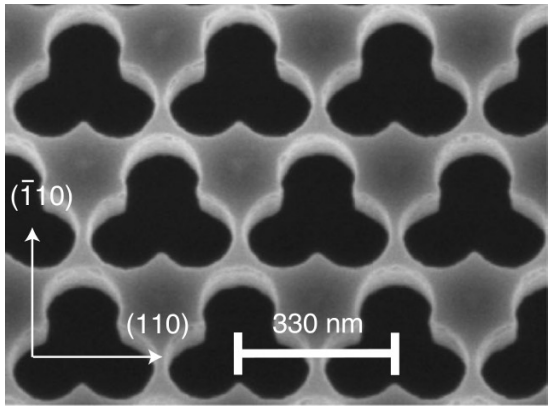
Several studies demonstrated changes in phonon dispersion cause by the holes:

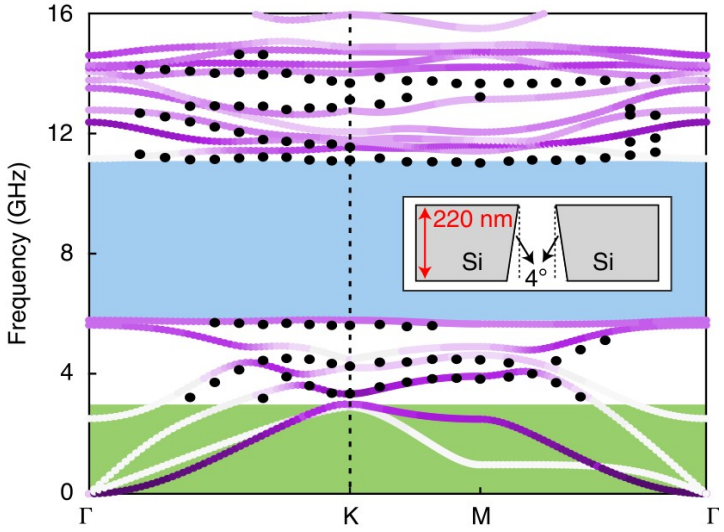
Graczykowski et al., PRB 91, 075414 (2015)



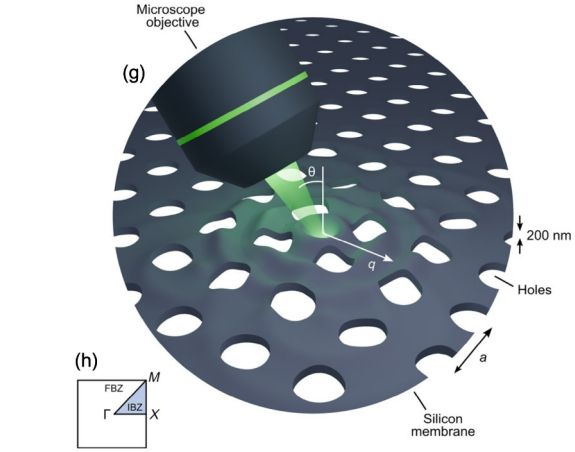


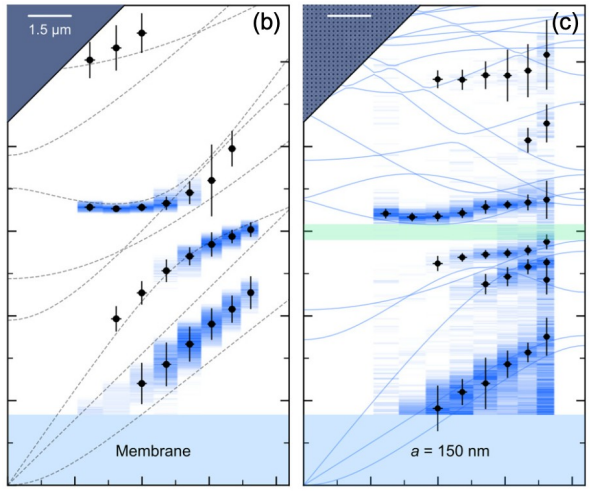
Florez et al., Nat. Nanotech. 17, 947 (2024)





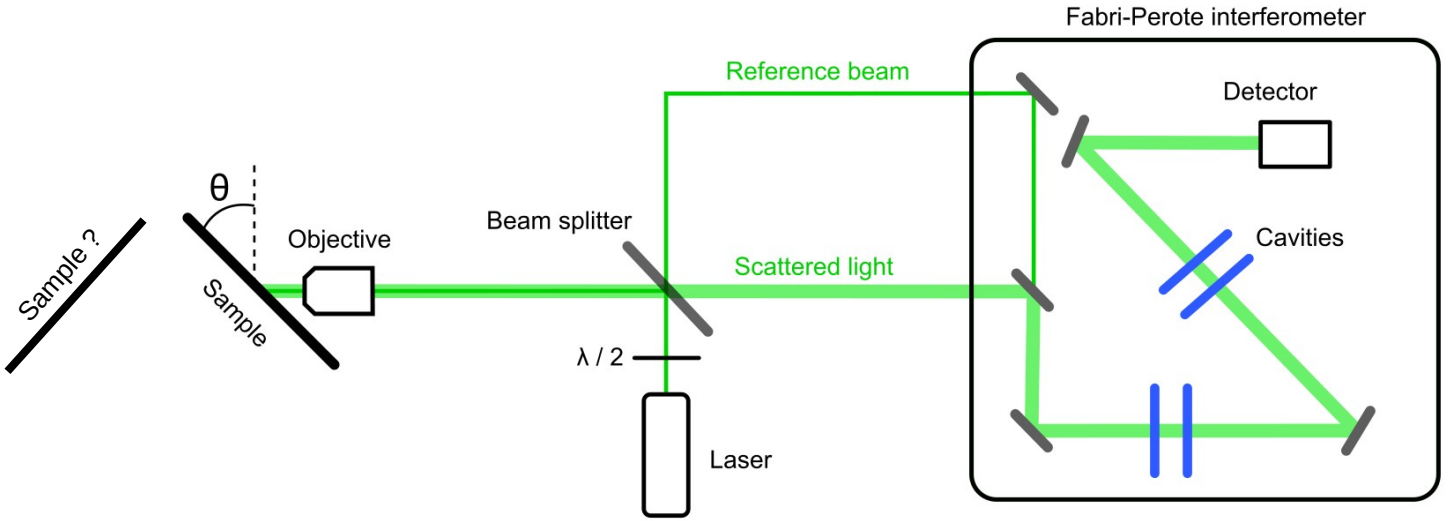
Anufriev et al., PRApplied (2025)

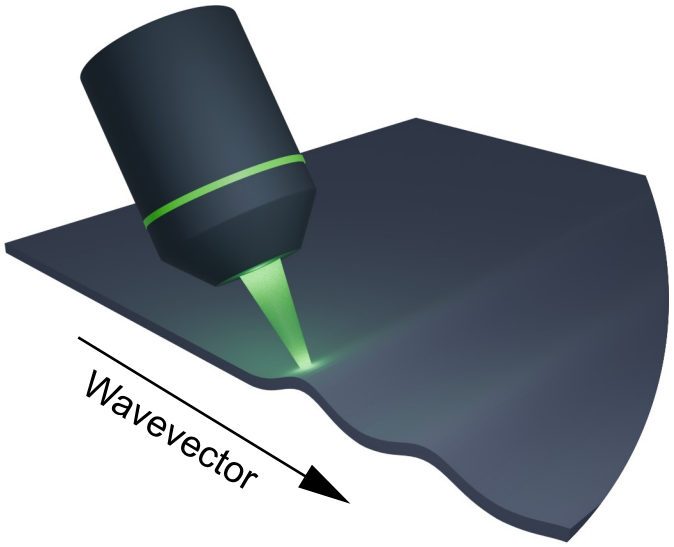


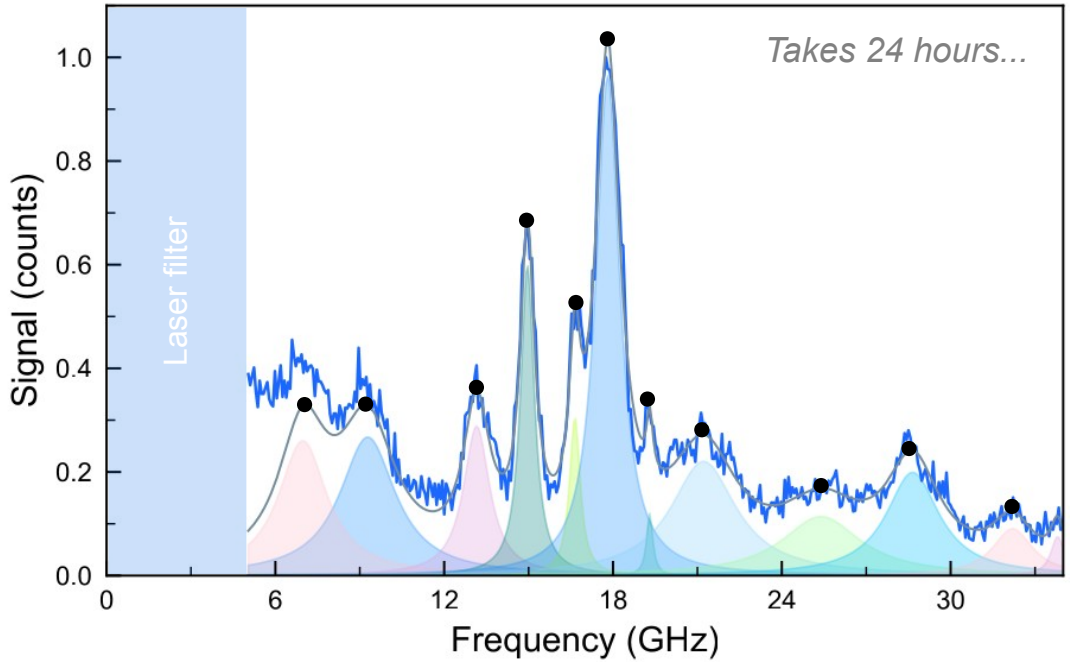


Brillouin-Mandelstam light scattering spectroscopy

The method enables measurements of phonon natural freqencies at a given wavevector:

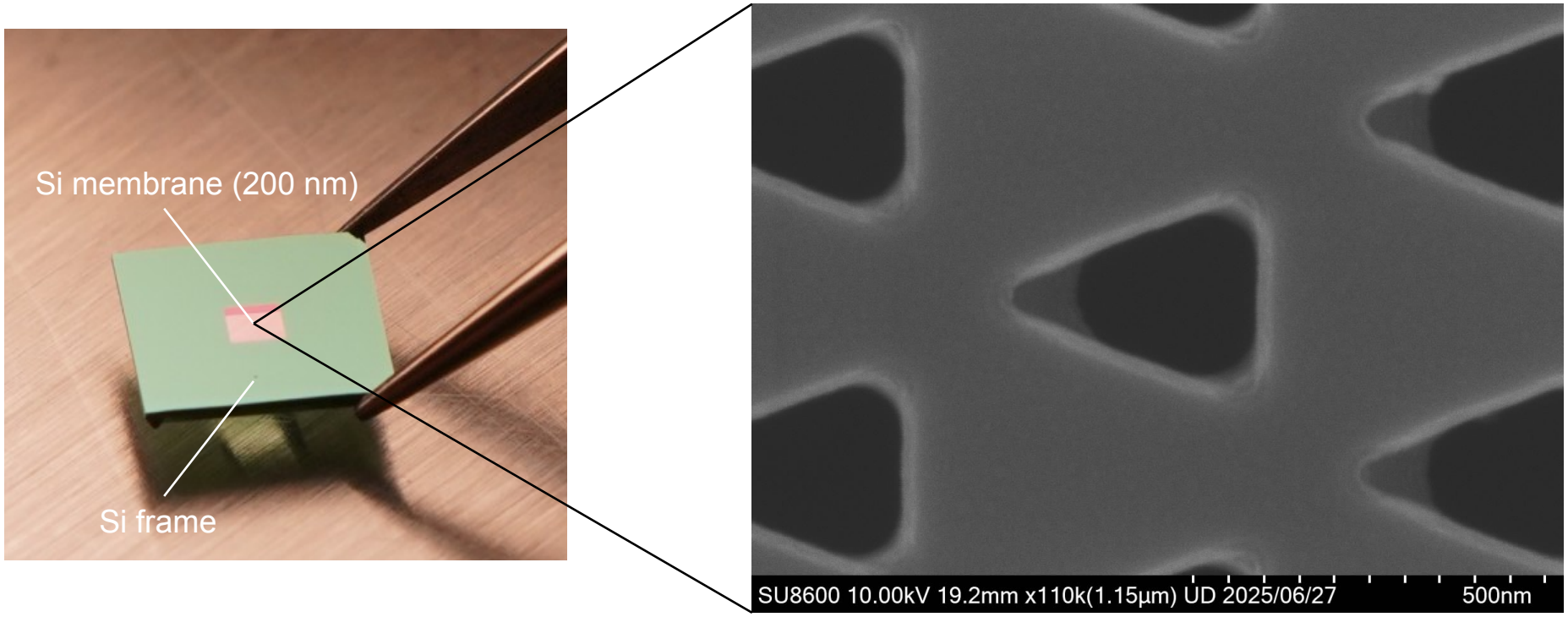


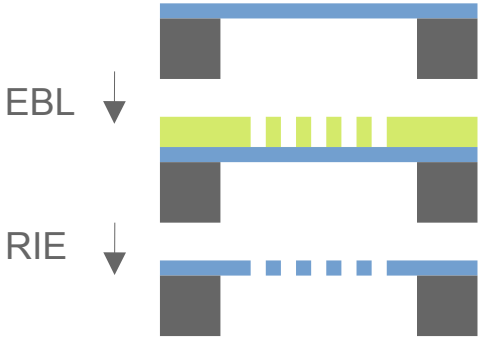




Samples

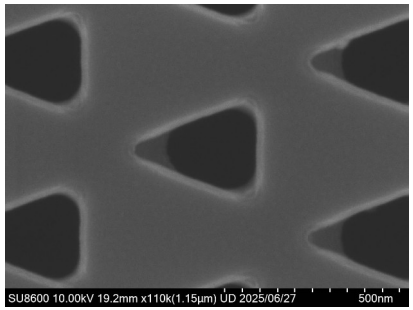
A 100 × 70 μm region with an array of triangular holes was fabricated:

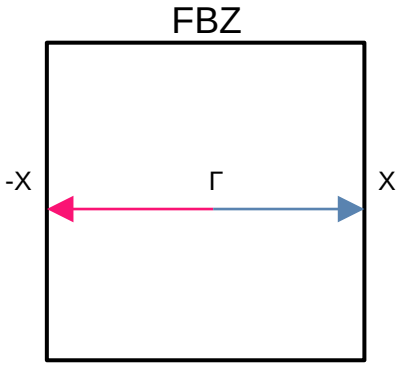


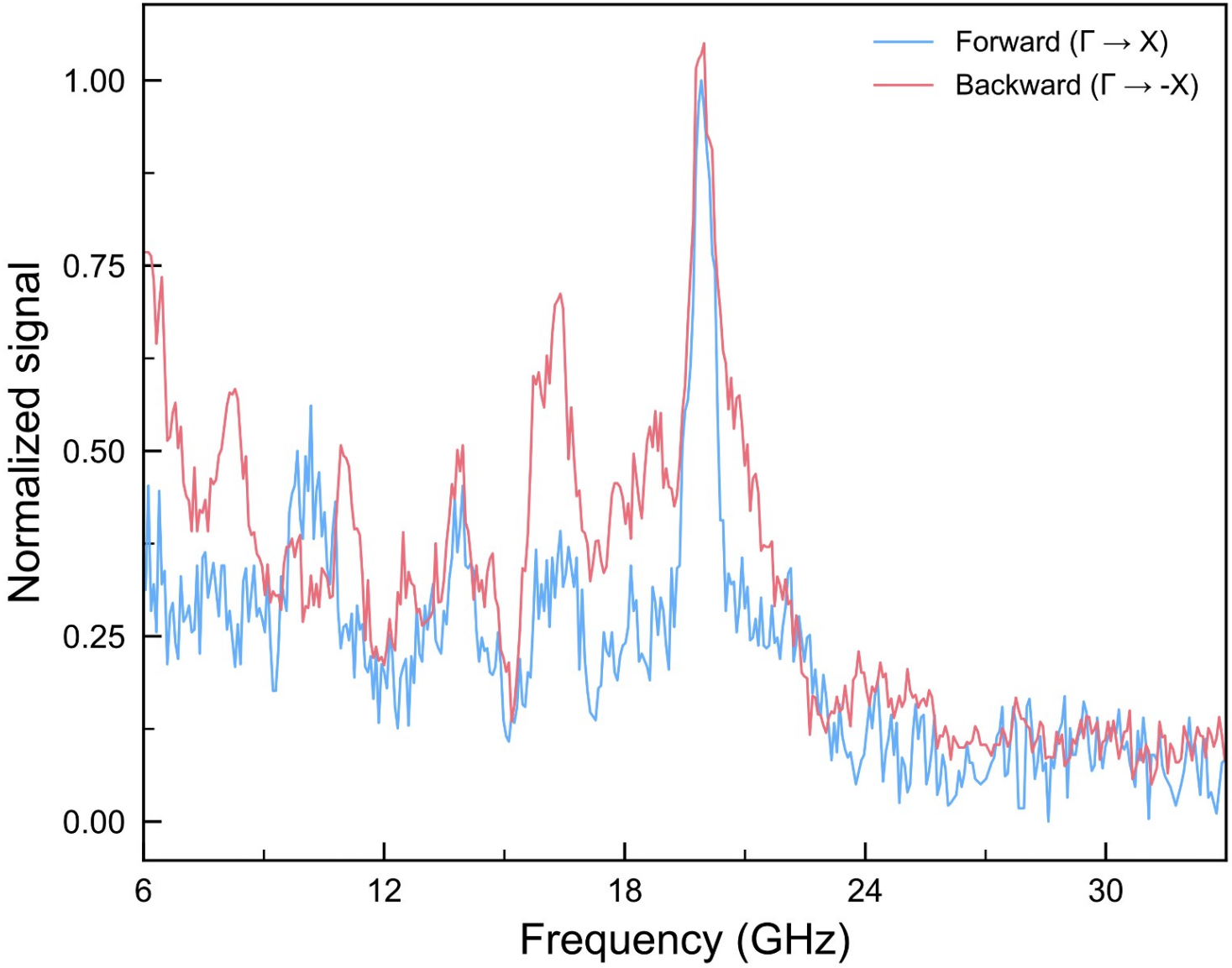


Spectra in different directions

This example shows a difference between spectra in the positive and negative directions:

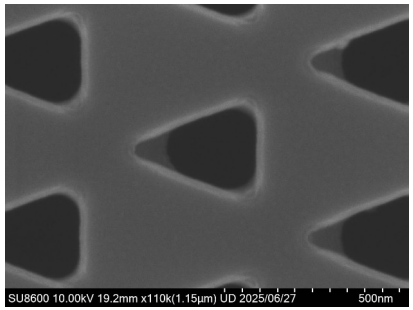


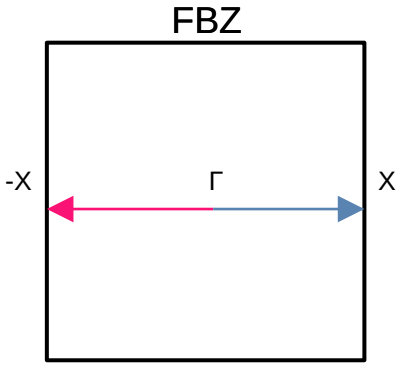


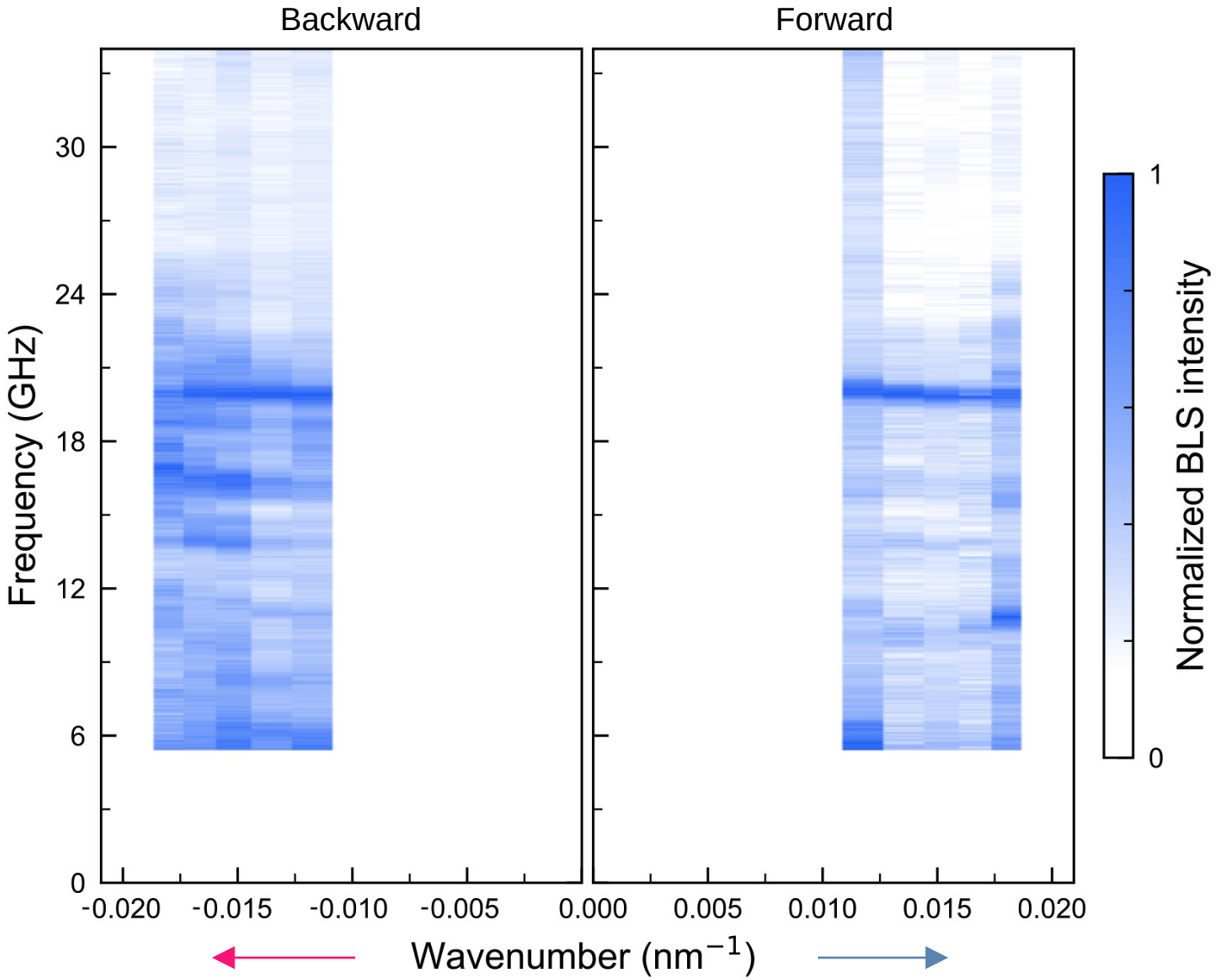


Spectra in different directions

The excited branches of the dispersion relation do not apper the same:







Conclusions

- We observe different phonon branches excited differently in the opposite directions
- This might explain thermal rectification and is likely caused by stronger backscattering from flat edges.
- New samples on Si and SiN are needed to confirm the effect